SOFTWARE ENGINEERING LAB

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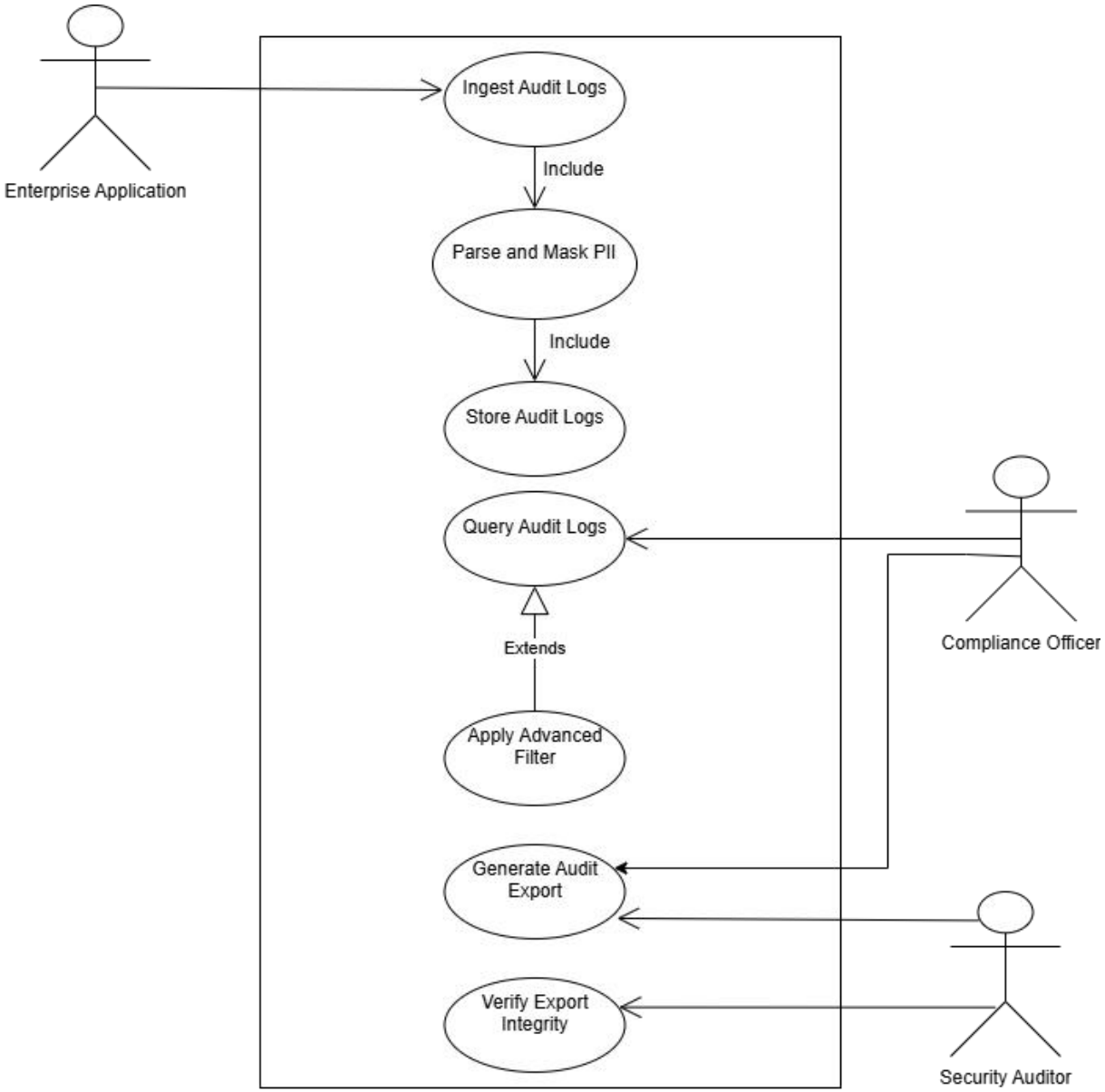
Scenario : 50-Centralized Audit Trail Compliance Engine

1.Requirements Table

Centralized Audit Trail Compliance Engine

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall parse structured JSON audit logs, identify and mask PII fields before persisting records.	High	Pass: A JSON audit log containing a supported PII field is stored with that field masked. Fail: PII remains in cleartext in the stored record.	Protects sensitive information before storage.
FR-002	Functional	The system shall allow enterprise application log sources to submit structured JSON audit logs.	High	Pass: A valid JSON audit log is accepted and processed. Fail: A valid log cannot be submitted.	Required to ingest enterprise audit events.
FR-003	Functional	The system shall allow authorized users to search and filter stored audit logs using supported criteria.	High	Pass: An authorized user can retrieve records matching the selected filter criteria.	Enables auditors to locate relevant audit records.
FR-004	Functional	The system shall allow authorized users to generate an audit export from selected audit records.	High	Pass: The system generates an export containing all selected records.	Provides audit evidence for compliance activities.
FR-005	Functional	The system shall allow an authorized Security Auditor to verify the integrity of an audit export.	High	Pass: An unmodified export passes verification and a modified export fails verification.	Detects unauthorized modification of audit evidence.
NFR-001	Nonfunctional	The system shall support complex filtering across 10 million historical records in under 1 second.	High	Pass: Benchmark tests show query response time below 1 second under the specified dataset/load.	Ensures efficient access to large audit histories.
NFR-002	Nonfunctional	The system shall ensure that generated audit exports contain cryptographic integrity information that allows modifications to be detected.	High	Pass: Modifying an exported record causes integrity verification to fail.	Supports cryptographically verifiable audit exports.

2. UML Use-Case Diagram



3. Use-Case Flow Document

For Use Case Generate Audit Export

Preconditions

1. The user is authenticated.
2. The user is authorized to generate audit exports.
3. Audit records are available in the system.

Main Success Scenario

1. The user selects "Generate Audit Export".
2. The system displays the available audit-log filtering options.
3. The user specifies the required audit records or filtering criteria.
4. The system retrieves the matching audit records.
5. The system prepares the selected audit records for export.
6. The system generates the audit export.
7. The system generates cryptographic integrity information for the export.
8. The system makes the generated export available to the user.
9. The system confirms that the export was generated successfully.
10. The use case ends successfully.

Postconditions

1. The requested audit records have been exported.

2. The generated export contains cryptographic integrity information.

3. The export is available to the authorized user.

Alternate Flow

4a. The system finds no audit records matching the specified criteria.

4a1. The system displays a "No matching records found" message.

4a2. The system does not generate the requested audit export.

4a3. The user may modify the filtering criteria and retry.

4a4. The use case ends.